



Required Capstone Project: Module 8

Due Jul 22 by 11:59pm**Points** None**Available** after Jul 15 at 12am☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459458>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459459>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459460>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459461>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459462>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459463>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459464>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459465>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459466>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459467>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459468>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459469>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459470>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459471>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459472>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459473>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459474>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459475>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459476>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459477>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459478>)☒  (<https://classroom.emeritus.org/courses/16839/modules/items/3459480>)☐  (<https://classroom.emeritus.org/courses/16839/modules/items/3459481>)☒  (<https://classroom.emeritus.org/courses/16839/modules/items/3459483>)

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🎯 Learning Outcomes Addressed

- Understand how AI, GenAI and Agentic AI work and where they apply in business.
- Use prompts effectively for analysis, decision-making and content generation.
- Map business workflows and identify where AI can drive automation and efficiency.
- Evaluate AI opportunities using feasibility, ROI and impact criteria.
- Build AI adoption roadmaps aligned to business goals and KPIs.
- Apply data governance, privacy and ethical AI principles.

Phase 3 – Agentic AI Design & Control

Milestone 8: Autonomous Workflow & Failure Design

Estimated Effort: 1.5 Hours

Objective

In the previous module, you designed a **Human-in-the-Loop governance structure** to ensure that AI agents operate safely with appropriate human oversight.

However, as organisations become more confident in AI systems, some workflows begin to operate with **higher levels of autonomy**.

In these cases:

- AI agents initiate and execute tasks automatically
- systems respond to **events or triggers** rather than manual requests
- workflows operate with **minimal human intervention**

Such systems are often described as **event-driven autonomous workflows**.

However, autonomous systems introduce new challenges. Organisations must ensure that these workflows remain:

- reliable
- observable
- controllable

In this module, you will design both:

- an **event-triggered autonomous workflow**, and
- a **failure-handling and monitoring strategy**

This ensures the AI system remains **trustworthy and resilient even when operating autonomously**.

How This Connects to Earlier Modules

Your submission should build on work developed in previous capstone milestones.

- **Module 5** – You redesigned the **AI-enabled future workflow**
- **Module 6** – You designed the **AI agent architecture**
- **Module 7** – You defined the **Human-in-the-Loop governance structure**

In this module, you will explore **what happens when parts of the workflow become autonomous**, and how the system manages **failures, errors, and monitoring**.

Note: This is a required assignment and counts towards programme completion.

Task Overview

Using the workflow and AI agent design from earlier modules, create a design for an **Autonomous Workflow & Failure Handling Strategy**.

Your design should explain:

- how workflows are **triggered automatically by events**
- how the agent **orchestrates multiple tools or systems**
- how the system detects **failures or incorrect outputs**
- how problems are **escalated to humans**
- how the system is **monitored for reliability**

This exercise helps you understand how **enterprise AI systems operate safely at scale**.

Step 1 — Define an Event-Triggered Agent Workflow

Autonomous workflows often begin with **events or triggers**.

An event may include:

- a new customer request
- a transaction being created
- a document uploaded to a system
- a threshold being exceeded in a monitoring system

Define what event **activates the AI agent** and describe the steps the agent performs once triggered.

Example (Illustrative)

Workflow: Invoice Processing

Event Trigger

A vendor uploads an invoice to the procurement system.

Autonomous Workflow

1. System detects new invoice upload
2. AI agent extracts invoice data
3. Agent validates invoice against purchase order records
4. If data matches, the system prepares the payment request
5. Finance team receives notification for final confirmation

This design shows how the system **reacts automatically to operational events**.

Step 2 — Define Tool Orchestration

AI agents rarely operate in isolation.

Instead, they coordinate across multiple enterprise tools and systems.

Define which tools your agent must interact with and explain how the agent **orchestrates them during execution**.

Examples of enterprise tools include:

- CRM systems
- accounting software
- document management platforms
- internal knowledge bases
- communication tools

Example (Illustrative)

Tool Orchestration

Tool	Purpose
Procurement system	Detect new invoice uploads
Document parser	Extract invoice data
Accounting system	Validate invoice information
Payment system	Initiate payment approval process

The agent coordinates these tools to **complete the workflow automatically**.

Step 3 — Design Failure Simulation Scenarios

No enterprise AI system operates perfectly.

Your design should consider **possible failure scenarios**.

Failures may occur due to:

- incomplete or ambiguous input data
- incorrect AI outputs
- system integration issues
- unavailable external services

Identify **2–3 potential failure scenarios** relevant to your workflow.

Example (Illustrative)

Failure Scenario 1

Invoice document format cannot be parsed correctly.

Failure Scenario 2

AI incorrectly extracts invoice amount from document.

Failure Scenario 3

The accounting system API becomes temporarily unavailable.

Considering these scenarios helps ensure the system remains **resilient under real operating conditions**.

Step 4 — Define the Escalation Path

When failures occur, the system must determine **how to respond and who should be notified**.

Define how your workflow escalates issues to human operators.

Escalation may include:

- routing tasks to human reviewers
 - notifying operations teams
 - pausing automated processing
 - creating alerts for system administrators
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Example (Illustrative)

Escalation Rule

If the AI agent cannot validate the invoice against purchase orders after two attempts, the case is automatically **escalated to a finance analyst for manual review**.

This ensures that failures are **resolved by human operators when necessary**.

Step 5 — Design a Monitoring Strategy

Autonomous systems must be **continuously monitored** to detect problems and ensure reliable performance.

Monitoring systems help organisations track:

- system errors
- AI output quality

- workflow execution rates
- operational performance

Define how the organisation would monitor the AI workflow.

Example (Illustrative)

Monitoring Strategy

The organisation tracks:

- the number of invoices processed automatically
- failure rates in data extraction
- average processing time per invoice

Alerts are triggered if:

- failure rates exceed a defined threshold
- system processing delays increase significantly

Monitoring ensures the system remains **transparent and controllable**.

Required Output Structure

Your **Autonomous Workflow & Failure Design** should include the following sections.

1. Event-Triggered Autonomous Workflow

Describe the event that activates the AI agent and outline the workflow steps that follow.

2. Tool Orchestration

Explain which systems and tools the AI agent interacts with.

3. Failure Simulation Scenarios

Identify possible failure cases that could occur within the workflow.

4. Escalation Path

Define how the system handles failures and which human roles are notified.

5. Monitoring Strategy

Explain how the organisation monitors system performance and reliability.

Practical Tips

Focus on Realistic Enterprise Workflows

Autonomous AI systems typically operate in **structured operational environments**, such as:

- finance processes
- customer support systems
- IT operations
- supply chain management

Choose failure scenarios that are **credible in real organisations**.

Design for Resilience

Enterprise AI systems must be designed to be:

- **resilient** – capable of handling failures
 - **observable** – performance can be monitored
 - **controllable** – humans can intervene when needed
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Consider Operational Visibility

Autonomous systems must still provide **clear operational visibility**.

Teams should always be able to understand:

- what the system is doing
 - when failures occur
 - how the system is performing over time
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Avoid These Common Mistakes

- ✗ Assuming AI systems never fail
- ✗ Designing workflows with no monitoring or alerts
- ✗ Creating fully autonomous systems without escalation paths
- ✗ Ignoring integration failures between enterprise systems

Reliable AI systems must always be **designed for failure and recovery**.

Submission

Before submitting your **Autonomous Workflow & Failure Design**, confirm that you have included:

- ✓ Event-triggered autonomous workflow
- ✓ Tool orchestration explanation
- ✓ Failure simulation scenarios
- ✓ Escalation path for operational issues
- ✓ Monitoring strategy for system performance

Instructions:

- Use the same document that you downloaded in Module 1 and include your responses.
- We recommend you add additional text boxes or slides (if needed) for each milestone task to ensure that your responses are documented in a proper format.
- The document will gradually grow as you complete each capstone milestone. The final submission occurs at the end of Module 14.

Grading Rubric

Maximum Points: 15

Criteria	Full Points	0 Points
Event-Triggered Autonomous Workflow (3 points)	The submission defines an event-triggered autonomous workflow, including triggering events, agent actions, workflow progression, and execution outcomes (3 points)	Event-triggered autonomous workflow is not defined. (0 points)

Criteria	Full Points	0 Points
Tool Orchestration Explanation (3 points)	The submission explains how agents interact with enterprise systems, tools, APIs, databases, or external applications to execute workflow activities. (3 points)	Tool orchestration and system interactions are missing (0 points)
Failure Simulation Scenarios (3 points)	The submission explains how the autonomous system responds to workflow failures, uncertain outputs, missing information, or execution errors through failure scenarios. (3 points)	Failure simulation scenarios are missing. (0 points)
Escalation Path for Operational Issues (3 points)	The submission defines escalation paths for operational issues, including conditions that trigger escalation, responsible stakeholders, and escalation actions. (3 points)	Escalation paths for operational issues are missing. (0 points)
Monitoring Strategy for System Performance (3 points)	The submission defines mechanisms for monitoring autonomous system performance, including performance metrics, alerts, reporting, and continuous oversight processes. (3 points)	Monitoring strategy for system performance is missing (0 points)

Why this Step Matters

As organisations adopt agentic AI systems, workflows increasingly move towards **autonomous execution**.

However, autonomy introduces new responsibilities.

Organisations must ensure that AI systems are:

- **reliable**
- **monitored**
- **auditable**
- **recoverable when failures occur**

Designing systems that anticipate failures is a critical step in building **trustworthy enterprise AI infrastructure**.

In the next phase of the capstone, you will explore how organisations manage **AI risk, governance, and strategic deployment at scale**.